

Лабораторная работа №3

1. Настройка VLAN, 802.1Q и Native VLAN

Созданием VLAN на всех коммутаторах. В консоли каждого свитча:

```
enable
configure terminal
vlan 20
name VLAN20
exit
vlan 333
name VLAN333
exit
end
write memory
```

Настраиваем access-порты для компьютеров, на Свитч-3 (PC1, PC2), Свитч-4 (PC3, PC4) и Свитч-5 (PC5, PC6):

```
enable
configure terminal
interface GigabitEthernet1/0
switchport mode access
switchport access vlan 20
no shutdown
exit
interface GigabitEthernet1/1
switchport mode access
switchport access vlan 333
no shutdown
exit
end
write memory
```

Настраиваем trunk-порты (с Native VLAN 333)

На Свитч-1 :

```
enable
configure terminal
interface range GigabitEthernet0/0-3, GigabitEthernet1/0-3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 333
switchport trunk allowed vlan 20,333
no shutdown
exit
end
write memory
```

На Свитч-2:

```
enable
configure terminal
interface range GigabitEthernet0/0-3, GigabitEthernet1/0-3
switchport trunk encapsulation dot1q
switchport mode trunk
```

```
switchport trunk native vlan 333
switchport trunk allowed vlan 20,333
no shutdown
exit
end
write memory
```

На Свитч-3, Свитч-4 и Свитч-5:

```
enable
configure terminal
interface range GigabitEthernet0/0-3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 333
switchport trunk allowed vlan 20,333
no shutdown
exit
end
write memory
```

Настраиваем IP-адреса на компьютерах:

```
PC1> ip 192.168.20.1/24      (VLAN 20)
PC2> ip 192.168.33.1/24     (VLAN 333)
PC3> ip 192.168.20.2/24     (VLAN 20)
PC4> ip 192.168.33.2/24     (VLAN 333)
PC5> ip 192.168.20.3/24     (VLAN 20)
PC6> ip 192.168.33.3/24     (VLAN 333)
```

2. Проверка доступности (Ping):

Проверяем доступность внутри VLAN20:

PC1> ping 192.168.20.2

```
PC1> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=64 time=18.914 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=64 time=8.005 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=64 time=10.964 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=64 time=11.746 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=64 time=19.133 ms
```

На скриншоте отражены успешные ответы от PC3.

Проверяем доступность внутри VLAN333:

PC2> ping 192.168.33.2

PC2> ping 192.168.33.3

```
PC2> ping 192.168.33.2

84 bytes from 192.168.33.2 icmp_seq=1 ttl=64 time=9.353 ms
84 bytes from 192.168.33.2 icmp_seq=2 ttl=64 time=17.081 ms
84 bytes from 192.168.33.2 icmp_seq=3 ttl=64 time=11.216 ms
84 bytes from 192.168.33.2 icmp_seq=4 ttl=64 time=18.259 ms
84 bytes from 192.168.33.2 icmp_seq=5 ttl=64 time=15.032 ms

PC2> ping 192.168.33.3

84 bytes from 192.168.33.3 icmp_seq=1 ttl=64 time=20.286 ms
84 bytes from 192.168.33.3 icmp_seq=2 ttl=64 time=16.561 ms
84 bytes from 192.168.33.3 icmp_seq=3 ttl=64 time=9.173 ms
84 bytes from 192.168.33.3 icmp_seq=4 ttl=64 time=19.313 ms
84 bytes from 192.168.33.3 icmp_seq=5 ttl=64 time=12.458 ms
```

На скриншоте представлены успешные ответы от PC4 и PC6.

Посмотрим недоступность между разными VLAN:

```
PC1> ping 192.168.33.1
```

```
PC1> ping 192.168.33.2
```

```
PC1> ping 192.168.33.1
```

```
No gateway found
```

```
PC1> ping 192.168.33.2
```

```
No gateway found
```

```
PC4> ping 192.168.20.2
```

```
PC4> ping 192.168.20.3
```

```
PC4> ping 192.168.20.2
```

```
No gateway found
```

```
PC4> ping 192.168.20.3
```

```
No gateway found
```

Коммутация на уровне L2 изолирует VLAN друг от друга. Для связи между VLAN необходима маршрутизация (L3).

3. Перехват пакетов с тегами и без тегов

Захват на линке с VLAN333 (Native):

```
PC2> ping 192.168.33.2
```

```
PC2> ping 192.168.33.2 -c 5
```

```
84 bytes from 192.168.33.2 icmp_seq=1 ttl=64 time=8.464 ms
```

```
84 bytes from 192.168.33.2 icmp_seq=2 ttl=64 time=22.010 ms
```

```
84 bytes from 192.168.33.2 icmp_seq=3 ttl=64 time=12.332 ms
```

```
84 bytes from 192.168.33.2 icmp_seq=4 ttl=64 time=9.521 ms
```

```
84 bytes from 192.168.33.2 icmp_seq=5 ttl=64 time=11.358 ms
```

Устанавливаем фильтр «icmp»:

The screenshot shows the Wireshark interface with a packet capture from 'Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]'. The filter is set to 'icmp'. The packet list shows 10 packets, including requests and replies. The packet details pane shows the structure of an Ethernet II frame, an IPv4 packet, and an ICMP Echo (ping) request.

No.	Time	Delta	Source	Destination	Protocol	Length	Info
17	12.943612		192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x7cc2, seq=1/256, ttl=64 (reply in 18)
18	12.949107	0.005495	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x7cc2, seq=1/256, ttl=64 (request in 17)
22	13.952785	1.003678	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x7dc2, seq=2/512, ttl=64 (reply in 23)
23	13.970772	0.017987	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x7dc2, seq=2/512, ttl=64 (request in 22)
24	14.974104	1.003332	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x7ec2, seq=3/768, ttl=64 (reply in 25)
25	14.985313	0.011209	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x7ec2, seq=3/768, ttl=64 (request in 24)
29	15.988378	1.003065	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x7fc2, seq=4/1024, ttl=64 (reply in 30)
30	15.995529	0.007151	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x7fc2, seq=4/1024, ttl=64 (request in 29)
31	16.998758	1.003229	192.168.33.1	192.168.33.2	ICMP	98	Echo (ping) request id=0x80c2, seq=5/1280, ttl=64 (reply in 32)
32	17.004253	0.005495	192.168.33.2	192.168.33.1	ICMP	98	Echo (ping) reply id=0x80c2, seq=5/1280, ttl=64 (request in 31)

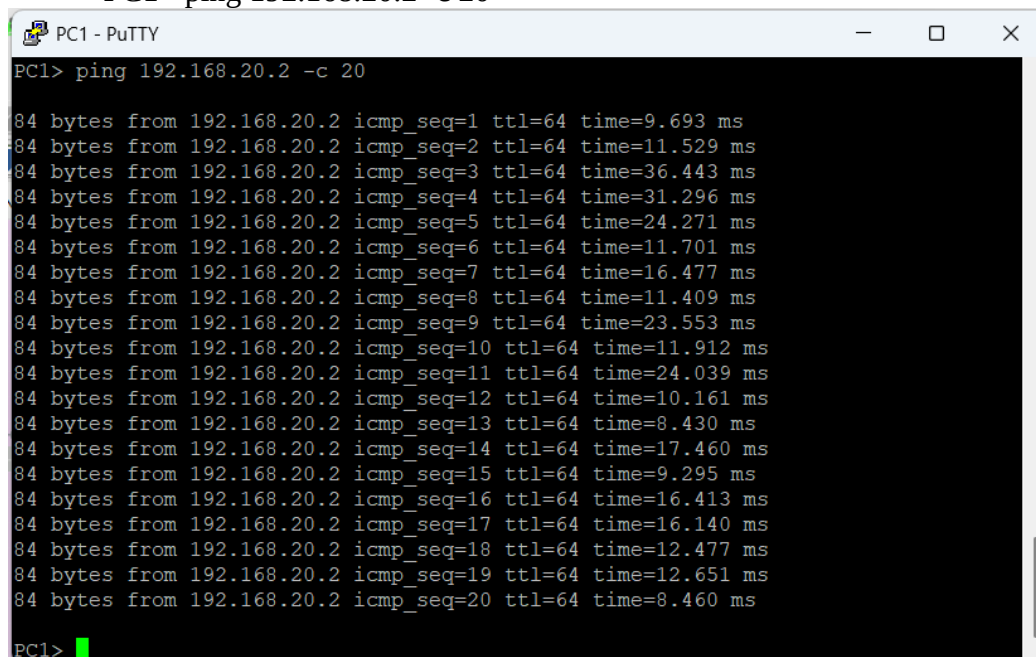
Frame 24: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0
Destination: Private_66:68:01 (00:50:79:66:68:01), Src: Private_66:68:03 (00:50:79:66:68:03)
Source: Private_66:68:01 (00:50:79:66:68:01)
Type: IPv4 (0x0800)
[Stream index: 5]
Internet Protocol Version 4, Src: 192.168.33.1, Dst: 192.168.33.2
Internet Control Message Protocol

На скриншоте показан ICMP-пакет Native VLAN 333. В заголовке кадра отсутствует тег 802.1Q, что соответствует настройке Native VLAN.

Захват выполнен на trunk-линке между SW2 и SW3 (порт Ethernet2). При фильтрации трафика с PC2 по адресу PC4 обнаружены ICMP-пакеты. В заголовке кадра отсутствует поле 802.1Q Virtual LAN. Это соответствует настройке Native VLAN 333, трафик которого передается без тега.

Захват на линке с VLAN20:

PC1> ping 192.168.20.2 -c 20

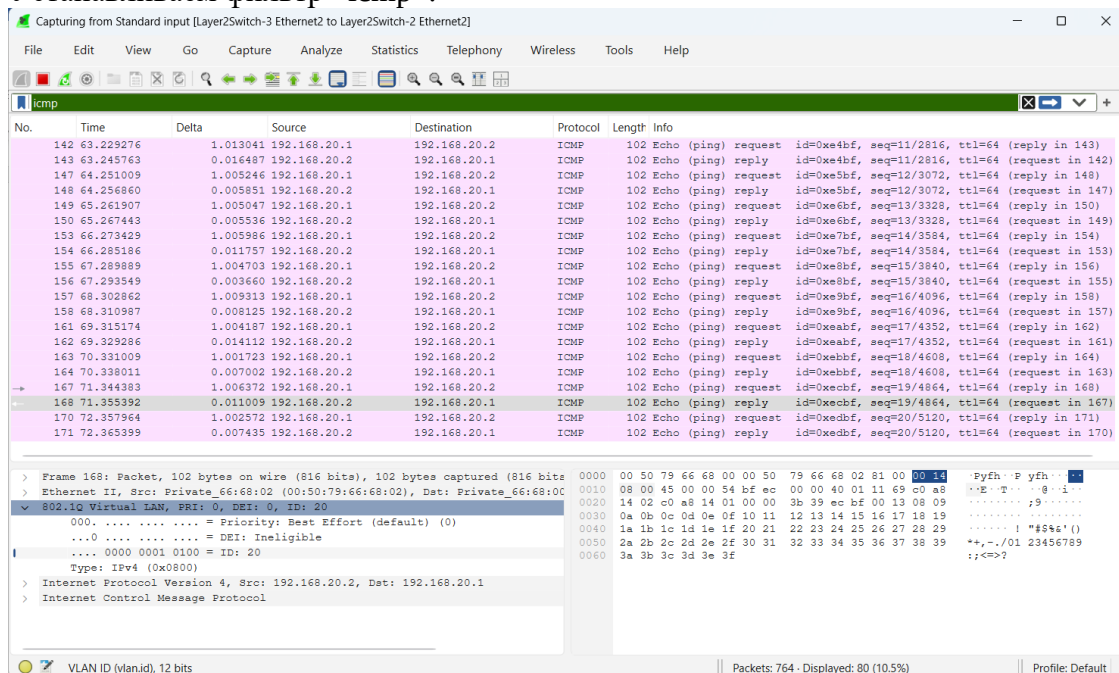


```
PC1> ping 192.168.20.2 -c 20

84 bytes from 192.168.20.2 icmp_seq=1 ttl=64 time=9.693 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=64 time=11.529 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=64 time=36.443 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=64 time=31.296 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=64 time=24.271 ms
84 bytes from 192.168.20.2 icmp_seq=6 ttl=64 time=11.701 ms
84 bytes from 192.168.20.2 icmp_seq=7 ttl=64 time=16.477 ms
84 bytes from 192.168.20.2 icmp_seq=8 ttl=64 time=11.409 ms
84 bytes from 192.168.20.2 icmp_seq=9 ttl=64 time=23.553 ms
84 bytes from 192.168.20.2 icmp_seq=10 ttl=64 time=11.912 ms
84 bytes from 192.168.20.2 icmp_seq=11 ttl=64 time=24.039 ms
84 bytes from 192.168.20.2 icmp_seq=12 ttl=64 time=10.161 ms
84 bytes from 192.168.20.2 icmp_seq=13 ttl=64 time=8.430 ms
84 bytes from 192.168.20.2 icmp_seq=14 ttl=64 time=17.460 ms
84 bytes from 192.168.20.2 icmp_seq=15 ttl=64 time=9.295 ms
84 bytes from 192.168.20.2 icmp_seq=16 ttl=64 time=16.413 ms
84 bytes from 192.168.20.2 icmp_seq=17 ttl=64 time=16.140 ms
84 bytes from 192.168.20.2 icmp_seq=18 ttl=64 time=12.477 ms
84 bytes from 192.168.20.2 icmp_seq=19 ttl=64 time=12.651 ms
84 bytes from 192.168.20.2 icmp_seq=20 ttl=64 time=8.460 ms

PC1>
```

Устанавливаем фильтр «icmp»:



No.	Time	Delta	Source	Destination	Protocol	Length	Info
142	63.229276	1.013041	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xe4bf, seq=11/2816, ttl=64 (reply in 143)
143	63.245763	0.016487	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe4bf, seq=11/2816, ttl=64 (request in 142)
147	64.251009	1.005246	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xe5bf, seq=12/3072, ttl=64 (reply in 148)
148	64.256860	0.005851	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe5bf, seq=12/3072, ttl=64 (request in 147)
149	65.261907	1.005047	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xe6bf, seq=13/3328, ttl=64 (reply in 150)
150	65.267443	0.005336	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe6bf, seq=13/3328, ttl=64 (request in 149)
153	66.273429	1.005966	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xe7bf, seq=14/3584, ttl=64 (reply in 154)
154	66.285186	0.011757	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe7bf, seq=14/3584, ttl=64 (request in 153)
155	67.289889	1.004703	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xe8bf, seq=15/3840, ttl=64 (reply in 156)
156	67.293549	0.003660	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe8bf, seq=15/3840, ttl=64 (request in 155)
157	68.302862	1.009313	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xe9bf, seq=16/4096, ttl=64 (reply in 158)
158	68.310987	0.008125	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe9bf, seq=16/4096, ttl=64 (request in 157)
161	69.315174	1.004187	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xeabf, seq=17/4352, ttl=64 (reply in 162)
162	69.329286	0.014112	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xeabf, seq=17/4352, ttl=64 (request in 161)
163	70.331009	1.001723	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xebbf, seq=18/4608, ttl=64 (reply in 164)
164	70.338011	0.007002	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xebbf, seq=18/4608, ttl=64 (request in 163)
167	71.344383	1.006372	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xecbf, seq=19/4864, ttl=64 (reply in 168)
168	71.355392	0.011009	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xecbf, seq=19/4864, ttl=64 (request in 167)
170	72.357964	1.002572	192.168.20.1	192.168.20.2	ICMP	102	Echo (ping) request id=0xedbf, seq=20/5120, ttl=64 (reply in 171)
171	72.365399	0.007435	192.168.20.2	192.168.20.1	ICMP	102	Echo (ping) reply id=0xedbf, seq=20/5120, ttl=64 (request in 170)

Frame 168: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface 0

Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: Private_66:68:0C:00:00:00 (00:50:79:66:68:0C:00:00:00)

0000 = Priority: Best Effort (default) (0)

0000 = DRI: Ineligible

..... 0000 0001 0100 = ID: 20

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 192.168.20.2, Dst: 192.168.20.1

Internet Control Message Protocol

На скриншоте показан ICMP-пакет, перехваченный на trunk-линке между SW2 и SW3. В заголовке кадра присутствует тег 802.1Q с VLAN ID 20, что подтверждает передачу тегированного трафика VLAN20.

Захват выполнен на trunk-линке между SW2 и SW3 (порт Ethernet2). При фильтрации трафика с PC1 по адресу PC3 обнаружены ICMP-пакеты. В заголовке кадра присутствует поле 802.1Q Virtual LAN. Значение VLAN ID: 20, что указывает на принадлежность к VLAN20. Присутствует тег 802.1Q (TPID = 0x8100).

4. Сохранение файлов конфигураций

В консоли всех свитчей вводим:

```
show running-config
```

Результат сохранен в файлы с именами, соответствующие именам устройств:

«Layer2Switch-1.txt», «Layer2Switch-2.txt», «Layer2Switch-3.txt», «Layer2Switch-4.txt» и «Layer2Switch-5.txt».